

CalmGuide

Development Report • 1 September 2026

This report covers a single continuous engagement on the CalmGuide codebase. It documents six workstreams: a private-testing invite-code gate, a full landing-page rebuild, a hands-on feature audit, a codebase cleanup and test-suite repair, read-aloud voice improvements, and a ten-phase UI redesign of the caregiver app.

Metric	Result
Commits shipped	13 — all pushed to origin/main
Files changed	173 (+4,208 / -8,619 lines)
Backend tests	1,584 passing / 1,585 (1 pre-existing, unrelated)
Frontend tests	242 passing, 1 skipped, 6 failing (started at 198 passing)
Type checking	tsc --noEmit clean
Production build	next build compiles successfully
Business logic changed	None — no API, schema, auth, RAG or SSE changes

1. Invite-Code Gate

A reusable invite-code system gating B2C signup during private testing. Codes are shared and reusable rather than single-use, so one code can be handed to a group of testers.

Backend

- New InviteCode model and Alembic migration (c5d6e7f8a9b0), storing an HMAC hash rather than the code itself.
- INVITE_CODE_REQUIRED settings toggle, enforced by default.
- Defence in depth: a client-side POST /invite-codes/validate check for fast feedback, plus a server-side re-check inside create_profile so the gate cannot be bypassed by calling the API directly.
- generate_invite_codes.py script for issuing codes out-of-band.
- 12 new tests covering rejection, acceptance, reusability, use-count increments, and that validation is read-only.

Frontend

- New StepInviteCode wizard step; profile setup went from 5 steps to 6.
- Existing test suite updated for the shifted step numbering.

Operational note — codes are not recoverable

Only the HMAC hash is stored, so generated codes are visible exactly once, at generation time. The plaintext codes live in access-codes.md, which is gitignored and never committed. There is currently no admin UI to list or revoke codes; deactivating one means setting is_active = false directly in the database.

2. Landing Page Rebuild

The public landing page was rebuilt from 16 loosely-grouped sections into a 12-section structure driven by a written brief, using the real CalmGuide and Leap of Faith logos and supplied product-screen imagery.

Structure

Nav → Hero → Value strip → What is CalmGuide → How it works → Moment Coach → Behavioral context → Safety → 3am experience → Healthcare / Technology → Final CTA → Footer.

Three previously standalone sections were folded into others rather than deleted: the RAG pipeline diagram became prose in “What is CalmGuide”, and the Voice section became a bullet in the 3am list. The DICE section had no slot in the new structure; its locale keys were left in place rather than deleted across all files.

Palette and identity

A supplied colour palette was applied. Notably, all eleven core brand colours already matched the specification exactly — verified programmatically rather than by eye — so this was an additive pass: three soft-tint values were adjusted, one new token added, and three gradients introduced (hero blob, final CTA, footer band).

Locale reduction

The frontend was reduced from 11 locales to 3 (English, Spanish, Hindi) — a product decision made earlier but not previously executed. This accounts for most of the deleted lines in this engagement.

3. Feature Audit

Every advertised feature was checked by hand against the running application — clicking through real flows, reading real AI responses, and inspecting the live database — rather than inferred from source code.

Verified working	Evidence
Invite-code gate	Live code accepted; invalid codes rejected
Profile setup wizard	Full 6-step flow completed end to end
AI Moment Coach	Real DICE-structured response, personalised, live LLM call
Caregiver check-in	Genuine empathetic reply, not canned
Learn / practice scenarios	Wrote a response, received real structured feedback
Incident auto-extraction	Coach conversation surfaced as an ABC card on Home
Pattern insights	Live frequency and trend data rendered on Home
Safety disclosure & crisis bar	Gate appears every time; 911/988/Helpline are real tel: links
Encryption at rest	Full write path round-trips through AES-256-GCM
Dark mode & 3 locales	Verified across app shell and landing page

Material finding — RAG is not actually retrieving anything

The local database contains no document or embedding table at all: the ingestion pipeline has never been run in this environment. Coach and scenario responses are genuinely good, but they run on the language model's own knowledge, not on retrieved trusted content. The landing page describes a “Trusted Knowledge (RAG)” pipeline, so this gap is worth closing before that claim is made publicly.

Not reached this pass: facility (B2B) screens beyond login (no seeded credentials in this environment), the daily behavioural log, manual incident entry, and confirmation that voice audio actually plays — the browser automation used cannot hear output.

4. Cleanup and Test-Suite Repair

Dead code removed

A full cross-reference sweep of every TypeScript file identified four components that had been present since the initial commit, referenced nowhere, and superseded by working equivalents. Nothing was removed on suspicion — each was verified unreferenced first.

Removed	Superseded by
BehavioralAnchorStage.tsx	StepDiseaseStage.tsx
PatternInsightsDetail.tsx	IncidentPatternCard.tsx
facility/TopBar.tsx	FacilityNav.tsx
facility/Sidebar.tsx	FacilityNav.tsx

Roughly a dozen orphaned locale keys left behind by these components and by the landing rebuild were also removed across all three locales. Two apparent dead files were investigated and correctly ruled out: `i18n/request.ts` (referenced as a string path by Next config) and `safety_redteam.py` (used by its test harness).

Test suite: 198 → 242 passing

Most failures traced to a single root cause: the global `next-intl` mock never implemented `t.has()`, cascading into roughly sixteen failures. The remainder was drift between refactored components and stale assertions — for example a chip component now rendering `role="radio"` while its test still looked for `role="option"`. All fixes were made to tests, tracing each component's actual current behaviour first; no component behaviour was changed to make a test pass.

Three items left deliberately unfixed

ThemeToggle (6 tests) — the button's accessible label describes the target state (“Switch to dark mode”) while the test expects the word “theme”. Which is correct is a content decision.

Card “elevated” variant — its style map is byte-for-byte identical to the default; no shadow is implemented. A genuine unimplemented feature, not a test bug.

StepChipSelector ARIA — wraps radio-role children in a `listbox` with `aria-multiselectable`, which is an invalid combination. Flagged rather than changed, as it touches real component markup.

5. Read-Aloud Voice

The read-aloud feature was using the browser's raw defaults: whatever voice happened to match the locale, at neutral rate and pitch, which reads flat and clinical — the opposite of what a caregiver at 3am needs.

- Added a curated allowlist of natural-sounding voices per platform, matched against what is actually installed, with graceful fallback.
- Slowed the rate to 0.95 and raised pitch to 1.05 for a calmer, warmer delivery.
- Fixed a latent bug: Chrome returns an empty voice list on first call until voices load asynchronously, so the very first read-aloud could silently fall back to a default voice. The list is now warmed on mount.
- Spanish was subsequently reported as still not friendly enough and switched from the classic “Mónica” to the newer, warmer “Reed” voice; English was left on “Samantha”, which was confirmed good.

Verified end to end through the real check-in flow in all three locales, confirming the intended voice, rate and pitch reached the speech engine. Note that audio output could not be heard directly — only that the correct parameters were applied.

6. UI Redesign — Ten Phases

The approved mobile design references were applied to the in-app caregiver experience across desktop and responsive mobile web. Visual system only.

Design tokens

As with the landing page, the existing palette already matched the brief exactly. The additions were four semantic Moment Coach section colours and a correction to dark-mode secondary text. A Tailwind colour namespace was deliberately *not* added, because text-coach already exists as a font-size token and would have collided.

Moment Coach response — the most visible change

The four response sections now carry distinct semantic tints so the hierarchy is readable at a glance under stress. All four exceed the WCAG AA threshold for normal text, not merely large text.

Section	Meaning	Contrast
Right Now (teal)	Act now	6.71 : 1
Why This Is Happening (lavender)	Context	9.39 : 1
What Not To Do (coral)	Avoid	6.56 : 1
When To Get More Help (navy)	Escalate	9.01 : 1

Desktop layout

The app shell previously rendered a 512-pixel column at every viewport width — the single largest reason desktop looked like stretched mobile. A new 1,040-pixel desktop width was introduced, deliberately narrower than the landing page's 1,200 so reading columns do not become uncomfortably long. Home now splits into two columns above the large breakpoint: the act-now path on the left, supporting context on the right. Because the source order already placed those sections sequentially, collapsing to one column reproduces the previous mobile order exactly.

Navigation

A mobile bottom navigation bar was added, mapped onto existing routes with no new pages: Coach, Insights, Resources, More. Nested routes keep their parent section active. It is hidden on desktop, on the landing page, and across facility routes. The references showed an inconsistent tab count between screens; four tabs were standardised everywhere rather than a bar that changes shape per screen.

Screen-level work

- **Incident Log** — all eight categories already mapped one-to-one to the backend enum, so this was presentation only. Single-letter placeholders (“A”, “?”, “W”) were replaced with real line icons in soft-tinted circles.
- **Daily Check-In** — icon-card severity selector and proper chips at 48px targets.
- **Care Profile** — discrete cards; calming strategies as teal chips and safety concerns as coral chips, mapping the references' “What Works Best / What to Avoid” pairing onto real fields.
- **Learn** — two-up card grid on wider screens. **Impact** — four-up stat row on desktop.
- **Facility** — consistency-only sweep onto the shared card class; density intentionally left alone.

Accessibility

Every visible interactive control on the redesigned screens was audited against the 44-pixel minimum and all failures fixed — all of them pre-existing. The toggle component's 40-pixel default was raised; the logo link and two Home links were effectively unhittable and are now properly sized. Toggle controls also gained `aria-pressed`; previously selection was conveyed by appending a check mark to the label text, which screen readers announced as part of the name.

7. Bugs Found and Fixed

Several of these were discovered incidentally and were not part of any requested scope.

Bug	Cause and fix
Logos invisible app-wide	Next's image optimizer sends <code>Content-Disposition: attachment</code> , which Chromium silently refuses to paint in an image element — no error, no failed request. Fixed by serving optimized images inline.
Backend would not start	The encryption key in <code>.env</code> was missing its base64 padding. Correcting it changed the secret used to hash invite codes, silently invalidating the first batch — caught by a failing live validation and regenerated.
Hydration mismatch on every page load	A pre-hydration script intentionally tags the document with theme and layout attributes the server cannot know. Missing <code>suppressHydrationWarning</code> .
Layout broke on language switch	The landing page's layout attribute was set once by a pre-hydration script that never re-ran on client-side navigation, so switching language brought back the app shell's chrome on top of the page. Fixed with an effect-based sync component.
Leap of Faith logo illegible	A wide horizontal lockup was being force-cropped into small circles, and rendered black-on-navy in the footer. Now used at natural aspect ratio with a light backing card.
Secondary button ignored theme	Its border and background were hardcoded to teal RGB values rather than the colour variable, so it silently failed to follow palette changes.

8. Open Items

Recommended before public launch

- **Run the RAG ingestion pipeline.** The “trusted knowledge” claim on the landing page is not currently backed by any ingested content in this environment.
- **Verify voice audio by ear.** Parameters are confirmed correct; the sound itself has not been independently judged.
- **Test facility (B2B) flows.** Everything past login remains unexercised for want of seeded credentials.
- **Decide the three deferred items** from section 4 — the toggle's label wording, the unimplemented card variant, and the invalid ARIA combination.

Known gaps

- **Two design references cannot be built as drawn.** Behavior Profile shows a photo, age, diagnosis and week-over-week stability metrics; Daily Check-In shows a five-point caregiver mood scale and loved-one state chips. None of those fields exist, and the patient's name is deliberately device-local and never sent to the server. Their visual language was adopted over real data instead of inventing the fields.
- **Desktop density is conservative.** Home is the only screen with a true two-column layout; focused single-task screens centre one column rather than inventing a second column of content.
- **Right-to-left support is dormant, not broken.** The check only fires for Arabic, which was removed in the locale reduction. The infrastructure is intact and nothing added here would break it.
- **The project has no ESLint configuration at all.** It has therefore never been linted. Adding one would surface a large volume of pre-existing issues and deserves to be its own task.

Appendix — Commit Log

Commit	Description
9e684d0	Add invite-code gate for private-testing signup
031822d	Rebuild landing page, reduce to 3 locales, misc fixes
18d6ac2	Fix landing-page chrome reappearing on client-side navigation
0d4107b	Audit pass: dead code cleanup, test suite repair, docs
4b3c97c	Make read-aloud sound friendly instead of a flat system voice
755fa5d	Swap Spanish read-aloud voice
a35a52a	UI redesign phases 1–3: tokens, Moment Coach sections, bottom nav
d5930c3	Desktop shell width, Home 2-column, Incident Log icons
121da0d	UI redesign phases 4–5 + design.md documentation
191dd88	UI redesign phases 7–9: Learn, Impact, facility consistency
f6f256c	UI redesign phase 10: touch-target accessibility pass

Commit	Description
ffce6f9	design.md: update known gaps after phases 7–10
761bb05	Gitignore .Rhistory editor artifact

Supporting documentation written during this engagement: design/design.md (design system, §25 covers this redesign) and docs/feature-audit-2026-09-01.md (the full feature audit).